Subspaces and quotients of L_1 defined by subsequences of the Haar system

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Overview

- Part I: Subspaces of L_1
- Part II: Quotients of L_1

Spans of subsequences of the Haar system

- Dyadic intervals: $\mathcal{D} = \{[0,1), [0,\frac{1}{2}), [\frac{1}{2},1), [0,\frac{1}{4}), [\frac{1}{4},\frac{1}{2}), \dots\}$
- Levels $\mathcal{D}^n = \{I \in \mathcal{D} : |I| = 2^{-n}\}$ for $n \in \mathbb{N}_0$
- Haar system: $h_I = \mathbb{1}_{I^+} - \mathbb{1}_{I^-}$, where I^+ and I^- are the left and right halves of I.
- Fix an infinite $\mathcal{A} \subset \mathcal{D}$ and $1 \le p < \infty$. Write

$$X = [h_I : I \in \mathcal{A}] \subset L_p[0,1] \quad \text{(closed span)}.$$

- Problem: determine the isomorphic type of X.

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Gamlen–Gaudet, Müller

- For $1 < p < \infty$, Gamlen and Gaudet (1973) classified the closed spans of subsequences of the Haar system in $L_p[0,1]$.
- $\limsup \mathcal{A} = \{t : t \in I \text{ for infinitely many } I \in \mathcal{A}\}$.
  $|\limsup \mathcal{A}| > 0 \implies X \simeq L_p$.
  $|\limsup \mathcal{A}| = 0 \implies X \simeq \ell_p$.
- Müller (1986) answered the question for dyadic H^1.
- Only three isomorphic types:

$$H^1, \qquad \ell_1, \qquad \left(\sum_{n=0}^{\infty} H_n^1\right)_{\ell_1}.$$

- H_n^1: the span in H^1 of the Haar functions up to level n.

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The Gamlen–Gaudet construction

If $|\limsup \mathcal{A}| > 0$:

- Select dyadic "building blocks" from $\mathcal{A}$.
- Form a block basis $(\hat{h}_I)$ of the Haar system that spans a complemented copy of L_p in $X = [h_I : I \in \mathcal{A}]$.

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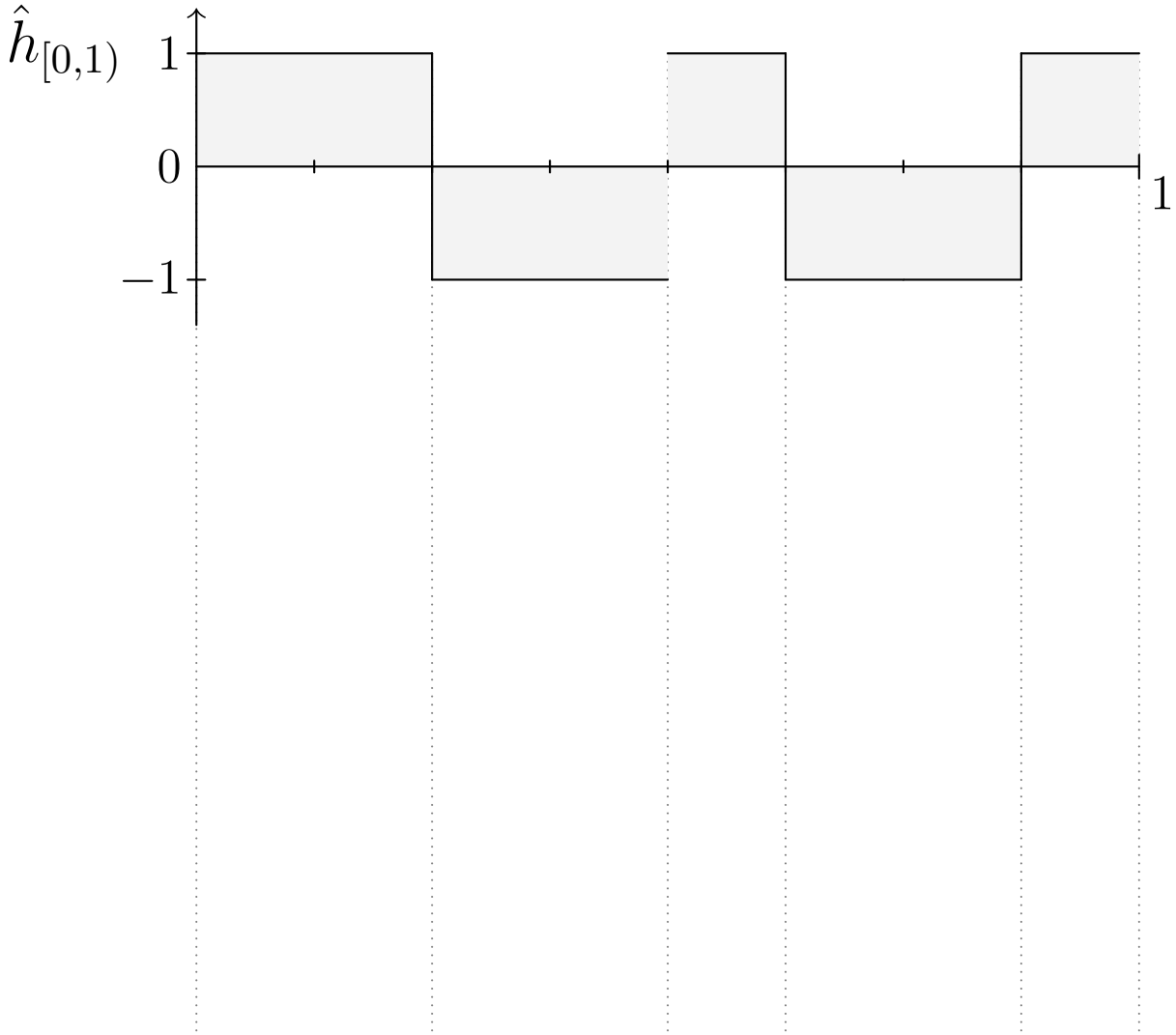
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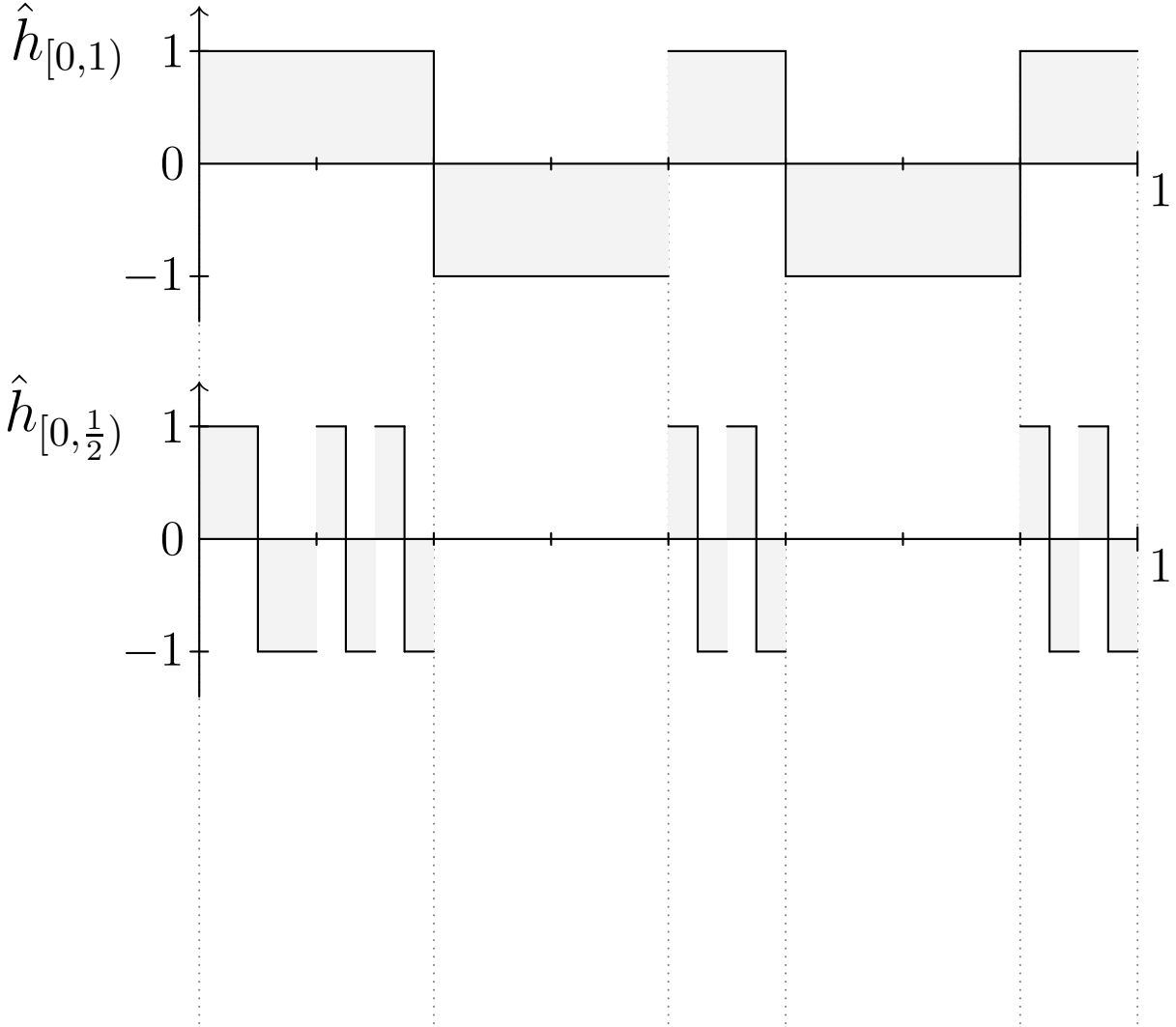
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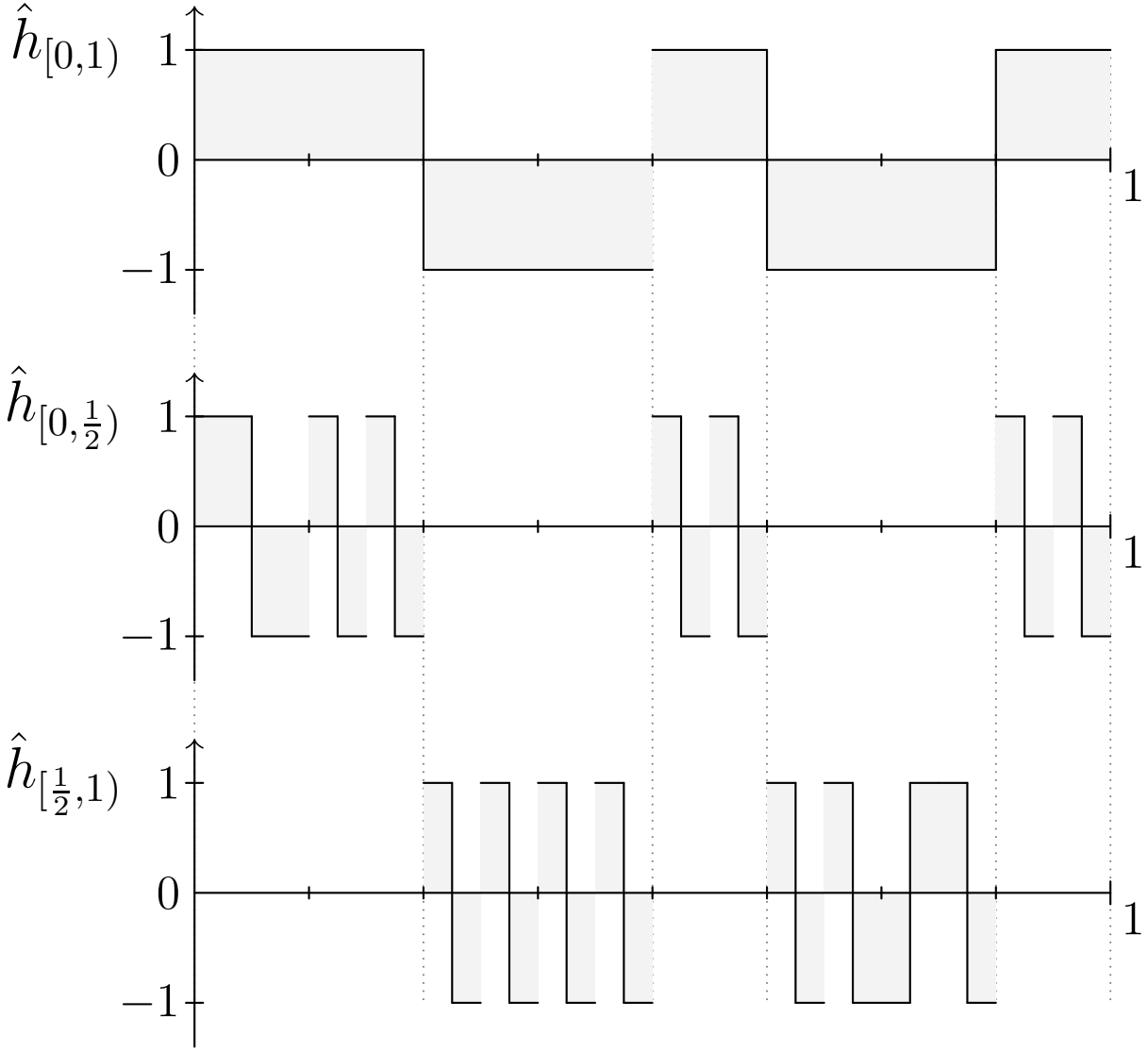
A block basis of the Haar system



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Pełczyński's decomposition method

Theorem (Pełczyński, 1960)

Suppose E and F embed complementably into one another and $E \simeq \ell_p(E)$ for some $1 \leq p < \infty$. Then $E \simeq F$.

- Gamlen–Gaudet: X contains a complemented subspace isomorphic to L_p.
- By unconditionality of the Haar basis for $1 < p < \infty$, X is complemented in L_p.
- Thus, $X \simeq L_p$.

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Why this argument fails in L_1

- When $p = 1$, the block construction still gives a subspace of X isomorphic to L_1 and complemented in X.
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The even and odd levels

$$X_0 = [h_I : I \in \mathcal{D}_{\text{even}}], \qquad X_1 = [h_I : I \in \mathcal{D}_{\text{odd}}].$$

- Are X_0 and X_1 isomorphic to L_1?
- Recorded by Martínez-Abejón–Odell–Popov (2005); attributed there to Schechtman.
- Independently asked by Dilworth and Kutzarova ($\to$ quasi-greedy basic sequences).

Theorem (S., 2026)

No.

Corollary: The cycle spaces $Z(D_n)$ of the diamond graphs are not uniformly isomorphic to $\ell_1^{\dim Z(D_n)}$ of the same dimension.

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Main result

Theorem (S., 2026)

Let X_0 and X_1 be the closed spans in $L_1[0,1]$ of the even, respectively odd, levels of the Haar system. Then

$$X_0 \not\simeq L_1[0,1] \qquad \text{and} \qquad X_1 \not\simeq L_1[0,1].$$

Proof.

- Construct $T\colon X_0 \to \ell_2$ bounded but not absolutely 2-summing.
- Every bounded $L_1 \to \ell_2$ operator is absolutely 2-summing.
- $X_0 \simeq X_1$, so we also obtain the conclusion for X_1.

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Absolutely 2-summing operators

- For a finite family $(x_i) \subset X$:

$$w_2(x_i) = \sup_{x^* \in B_{X^*}} \left(\sum_i |x^*(x_i)|^2 \right)^{1/2}.$$

- We call $T : X \to Y$ absolutely 2-summing if

$$\left(\sum_i \|Tx_i\|^2 \right)^{1/2} \leq C\, w_2(x_i) \quad \text{for every finite family } (x_i).$$

  The least such C is $\pi_2(T)$.

- A consequence of Grothendieck's inequality: $\pi_2(S) \leq K_G \|S\|$ for every bounded $S : L_1 \to \ell_2$.

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The operator T

$$T : X_0 \longrightarrow \ell_2(\mathcal{D}_{\text{even}}), \qquad Tf = \left(\int_0^1 f h_I \right)_{I \in \mathcal{D}_{\text{even}}}.$$

- Goal 1: $\|Tf\|_2 \leq \sqrt{2}\,\|f\|_1$.
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Odd-level cancellation

- Fix $I \in \mathcal{D}_{\text{even}}$; let I_1, I_2, I_3, I_4 be its consecutive quarters.
- For $f \in X_0$, the odd-level Haar coefficients on I^+ and I^- vanish:

$$\int_{I_1} f = \int_{I_2} f = \alpha, \qquad \int_{I_3} f = \int_{I_4} f = \beta.$$

- Hence

$$\int f h_I = 2(\alpha - \beta).$$

- Put $a_J = \int_J |f|$; then $a_{I_1}, a_{I_2} \geq |\alpha|$ and $a_{I_3}, a_{I_4} \geq |\beta|$.

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Local square estimate

- Put $a_J = \int_J |f|$; then $a_{I_1}, a_{I_2} \geq |\alpha|$ and $a_{I_3}, a_{I_4} \geq |\beta|$.

$$a_I^2 - \sum_{j=1}^{4} a_{I_j}^2 = 2 \sum_{1 \leq j < k \leq 4} a_{I_j} a_{I_k}$$

$$\geq 2(|\alpha|^2 + |\beta|^2 + 4|\alpha||\beta|)$$

$$\geq 2|\alpha - \beta|^2$$

$$= \frac{1}{2} \left| \int f h_I \right|^2 .$$

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Telescoping over even levels

- The quarters of the intervals in $\mathcal{D}^{2n}$ are exactly the intervals in $\mathcal{D}^{2n+2}$.
- We get a telescoping sum:

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A weakly 2-bounded test family

- $e_I = |I|^{-1/2} h_I$, $I \in \mathcal{D}_{\text{even}}$: orthonormal in L_2.
- For finite $\mathcal{C} \subset \mathcal{D}_{\text{even}}$ and $x^* \in B_{X_0^*}$, Hahn–Banach gives $g \in B_{L_\infty}$ with $x^*(e_I) = \int g e_I$.
- Bessel's inequality:

$$\sum_{I \in \mathcal{C}} |x^*(e_I)|^2 \leq \|g\|_2^2 \leq \|g\|_\infty^2 \leq 1.$$

- Thus $w_2(e_I : I \in \mathcal{C}) \leq 1$, independently of $\mathcal{C}$.

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The images grow without bound

- Let δ_I be the Ith unit vector of $\ell_2(\mathcal{D}_{\text{even}})$. Then

$$Te_I = |I|^{1/2}\delta_I.$$

- Let $\mathcal{C}_N = \bigcup_{n=0}^{N} \mathcal{D}^{2n}$. Then

$$\sum_{I\in\mathcal{C}_N} \|Te_I\|_2^2 = \sum_{n=0}^{N} \sum_{I\in\mathcal{D}^{2n}} |I| = N+1.$$

- $w_2(e_I : I \in \mathcal{C}_N) \leq 1$, but $\left(\sum_{I\in\mathcal{C}_N} \|Te_I\|_2^2\right)^{1/2} = \sqrt{N+1}$.
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X_0 and X_1 are isomorphic

- Observe that

$$X_1 \simeq X_0 \oplus X_0 \quad \text{and} \quad X_0 \simeq \mathbb{R} \oplus X_1 \oplus X_1.$$

- Moreover, $X_0 \simeq \ell_1(X_0)$.
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Part II

Quotients of L_1

Joint work with Stephen Dilworth, Denka Kutzarova, and Mikhail Ostrovskii

Quotients by Haar subsequences

- Fix any $\mathcal{A} \subset \mathcal{D}$, let $Y = [h_I : I \in \mathcal{A}] \subset L_1$, and consider

$$L_1/Y, \qquad \|f + Y\|_{L_1/Y} = \inf_{y \in Y} \|f + y\|_1.$$

- A Banach space X has cotype 2 if, for every finite family (x_j) in X,

$$\left(\sum_j \|x_j\|^2\right)^{1/2} \leq C\, \mathbb{E}_\varepsilon \left\|\sum_j \varepsilon_j x_j\right\|.$$

  Here ε_j are independent random signs and C is fixed.

- L_1 has cotype 2 (by Minkowski's and Khintchine's inequalities).
- Cotype 2 passes to subspaces, but not to arbitrary quotients.

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Transportation cost spaces on diamond graphs

- For a (finite, connected) graph G with its shortest-path metric,

$$\mathrm{TC}(G) \equiv \ell_1(E(G))/Z(G),$$

  where $E(G)$ is the edge set and $Z(G)$ is the cycle space.

- For the diamond graphs D_n, $\ell_1(E(D_n))$ corresponds to the span of the first $2n$ Haar levels (and $\mathbb{1}$) in L_1.

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The cotype result

Theorem (Dilworth–Kutzarova–Ostrovskii–S., 2026)

Let $\mathcal{A} \subset \mathcal{D}$, and let Y denote the closed linear span of $\{h_I : I \in \mathcal{A}\}$ in L_1. Then the quotient space L_1/Y has cotype 2.

- In particular, this holds for L_1/X_0 and L_1/X_1.
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Finite-dimensional reduction

- Work in the finite-dimensional space

$$L_1^N = \operatorname{span}\{\mathbb{1}, h_I : I \in \mathcal{D}_N\}, \qquad \mathcal{D}_N = \{I \in \mathcal{D} : |I| \geq 2^{-N}\}.$$

- Fix $\mathcal{A} \subset \mathcal{D}_N$ and put

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- For $x_1, \ldots, x_n \in L_1^N$, prove

$$\Big(\sum_{j=1}^n \|x_j + Y\|_{L_1/Y}^2\Big)^{1/2} \leq C\,\mathbb{E}_\varepsilon \Big\|\sum_{j=1}^n \varepsilon_j x_j + Y\Big\|_{L_1/Y},$$

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The target inequality and minimizing representatives

For $x_1, \ldots, x_n \in L_1^N$, we want

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$$F_\varepsilon = \sum_j \varepsilon_j x_j + y_\varepsilon, \quad y_\varepsilon \in Y, \qquad \|F_\varepsilon\|_{L_1} = \left\|\sum_j \varepsilon_j x_j + Y\right\|_{L_1/Y}.$$

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A linear choice of representatives

- For a positive weight $w \in L_1^N$, put

$$\langle f, g \rangle_w = \int \frac{fg}{w},$$

and let P_w be the orthogonal projection onto $Y^{\perp_w}$.

- Since $P_w|_Y = 0$ and $P_w f - f \in Y$, the vectors

$$z_j = P_w x_j$$

give "good" representatives of $x_j + Y$ by a linear rule.

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- We will use
$$w = \mathbb{E}_\varepsilon |F_\varepsilon| + \eta, \qquad \eta > 0.$$
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$$\left(\sum_j \|x_j + Y\|_{L_1/Y}^2\right)^{1/2} \leq \int \left(\sum_j |z_j|^2\right)^{1/2} \lesssim \int \left(\mathbb{E}_\varepsilon \left|\sum_j \varepsilon_j z_j\right|^{1/2}\right)^2.$$
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Theorem (Bourgain–Davis, 1986, $p = 1/2$)

For a martingale transform T on a finite filtered measure space (Ω, ν), with predictable multipliers of modulus at most 1,

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In mixed-norm notation, $\|Th\|_{L_2(\nu;L_{1/2}(S))} \lesssim \|h\|_{L_2(\nu;L_1(S))}$, where $S = \{1, \ldots, m\}$ has the uniform probability measure and $h(t,i) = f_i(t)$ (T acts in the variable t).

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In mixed-norm notation, $\|Th\|_{L_2(\nu;L_{1/2}(S))} \lesssim \|h\|_{L_2(\nu;L_1(S))}$, where $S = \{1, \ldots, m\}$ has the uniform probability measure and $h(t,i) = f_i(t)$ (T acts in the variable t).

The Bourgain–Davis inequality

$$\int \left(\mathbb{E}_\varepsilon |P_w F_\varepsilon|^{1/2}\right)^4 \frac{dt}{w} \lesssim \int \left(\mathbb{E}_\varepsilon |F_\varepsilon|\right)^2 \frac{dt}{w}$$

Theorem (Bourgain–Davis, 1986, $p = 1/2$)

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The function for the change of measure

- To obtain the missing estimate, we identify P_w as a martingale transform after a change of measure.
- Define

$$b = P_w \mathbb{1}.$$

  It satisfies

$$b - \mathbb{1} \in Y, \qquad \langle b, y \rangle_w = 0 \quad (y \in Y).$$

- One can show that b is strictly positive on $[0, 1)$.

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The new measure and its Haar system

- Define the positive measure ν and the ν-Haar functions by

$$d\nu = \frac{b^2}{w}\,dt, \qquad h_I^\nu = \frac{\mathbb{1}_{I^+}}{\nu(I^+)} - \frac{\mathbb{1}_{I^-}}{\nu(I^-)}.$$

- Multiplication by b preserves the inner products:

$$\langle bg, bh\rangle_w = \int gh\,d\nu.$$

- The Haar structure also gives

$$Y = \operatorname{span}\{bh_I^\nu : I \in \mathcal{A}\}.$$

These functions form a weighted orthogonal basis of Y.

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The projection as a martingale transform

- Let T_ν keep the constant and delete precisely the selected ν-Haar coefficients:

$$T_\nu \mathbb{1} = \mathbb{1}, \qquad T_\nu h_I^\nu = \begin{cases} 0, & I \in \mathcal{A}, \\ h_I^\nu, & I \notin \mathcal{A}. \end{cases}$$

- Multiplication by b identifies the deleted subspace with Y and preserves orthogonality. Thus

$$\boxed{P_w f = b\, T_\nu (f/b).}$$

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Thank you for your attention!